2. Binary Operations

1. $b * d = e, \quad c * c = b, \quad [(a * c) * e] * a = [c * e] * a = a * a = a$

2. $(a * b) * c = b * c = a$ and $a * (b * c) = a * a = a$, so the operation might be associative, but we can't tell without checking all other triple products.

3. $(b * d) * c = e * c = a$ and $b * (d * c) = b * b = c$, so the operation is not associative.

4. It is not commutative because $b * e = c$ but $e * b = b$.

5. Now $d * a = d$ so fill in d for $a * d$. Also, $c * b = a$ so fill in a for $b * c$. Now $b * d = c$ so fill in c for $d * b$. Finally, $c * d = b$ so fill in b for $d * c$.

6. $d * a = (c * b) * a = c * (b * a) = c * b = d$. In a similar fashion, substituting $c * b$ for d and using the associative property, we find that $d * b = c, d * c = c$, and $d * d = d$.

7. It is not commutative because $1-2 \neq 2-1$. It is not associative because $2 = 1-(2-3) \neq (1-2)-3 = -4$.

8. It is commutative because $ab + 1 = ba + 1$ for all $a, b \in \mathbb{Q}$. It is not associative because $(a * b) * c = (ab + 1) * c = abc + c + 1$ but $a * (b * c) = a * (bc + 1) = abc + a + 1$, and we need not have $a = c$.

9. It is commutative because $ab/2 = ba/2$ for all $a, b \in \mathbb{Q}$. It is associative because $a*(b*c) = a*(bc/2) = [a(bc/2)]/2 = abc/4$, and $(a * b) * c = (ab/2) * c = [(ab/2)c]/2 = abc/4$ also.

10. It is commutative because $2^{ab} = 2^{ba}$ for all $a, b \in \mathbb{Z}^+$. It is not associative because $(a*b)*c = 2^{ab} * c = 2^{(2^{ab})c}$, but $a * (b * c) = a * 2^{bc} = 2^{a(2^{bc})}$.

11. It is not commutative because $2 * 3 = 2^3 = 8 \neq 9 = 3^2 = 3 * 2$. It is not associative because $a * (b * c) = a * b^c = a^{(b^c)}$, but $(a * b) * c = a^b * c = (a^b)^c = a^{bc}$, and $bc \neq b^c$ for some $b, c \in \mathbb{Z}^+$.

12. If S has just one element, there is only one possible binary operation on S; the table must be filled in with that single element. If S has two elements, there are 16 possible operations, for there are four places to fill in a table, and each may be filled in two ways, and $2 \cdot 2 \cdot 2 \cdot 2 = 16$. There are 19,683 operations on a set S with three elements, for there are nine places to fill in a table, and $3^9 = 19,683$. With n elements, there are n^2 places to fill in a table, each of which can be done in n ways, so there are $n^{(n^2)}$ possible tables.

13. A commutative binary operation on a set with n elements is completely determined by the elements on or above the *main diagonal* in its table, which runs from the upper left corner to the lower right corner. The number of such places to fill in is

$$n + \frac{n^2 - n}{2} = \frac{n^2 + n}{2}.$$

Thus there are $n^{(n^2+n)/2}$ possible commutative binary operations on an n-element set. For $n = 2$, we obtain $2^3 = 8$, and for $n = 3$ we obtain $3^6 = 729$.

14. It is incorrect. Mention should be made of the underlying set for $*$ and the universal quantifier, *for all*, should appear.

A binary operation $*$ on a set S is **commutative** if and only if $a * b = b * a$ for all $a, b \in S$.

15. The definition is correct.

16. It is incorrect. Replace the final S by H.

17. It is not a binary operation. Condition 2 is violated, for $1 * 1 = 0$ and $0 \notin \mathbb{Z}^+$.

18. This does define a binary operation.

19. This does define a binary operation.

20. This does define a binary operation.

21. It is not a binary operation. Condition 1 is violated, for $2 * 3$ might be any integer greater than 9.

22. It is not a binary operation. Condition 2 is violated, for $1 * 1 = 0$ and $0 \notin \mathbb{Z}^+$.

23. **a.** Yes. $\begin{bmatrix} a & -b \\ b & a \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} a+c & -(b+d) \\ b+d & a+c \end{bmatrix}$.

b. Yes. $\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} ac-bd & -(ad+bc) \\ ad+bc & ac-bd \end{bmatrix}$.

24. F T F F F T T T T F **25.** (See the answer in the text.)

26. We have $(a*b)*(c*d) = (c*d)*(a*b) = (d*c)*(a*b) = [(d*c)*a]*b$, where we used commutativity for the first two steps and associativity for the last.

27. The statement is true. Commutativity and associativity assert the equality of certain computations. For a binary operation on a set with just one element, that element is the result of every computation involving the operation, so the operation must be commutative and associative.

28.

$*$	a	b
a	b	a
b	a	a

The statement is false. Consider the operation on $\{a, b\}$ defined by the table. Then $(a * a) * b = b * b = a$ but $a * (a * b) = a * a = b$.

29. It is associative.
Proof: $[(f + g) + h](x) = (f + g)(x) + h(x) = [f(x) + g(x)] + h(x) = f(x) + [g(x) + h(x)] = f(x) + [(g + h)(x)] = [f + (g + h)](x)$ because addition in $\mathbb{R}$ is associative.

30. It is not commutative. Let $f(x) = 2x$ and $g(x) = 5x$. Then $(f-g)(x) = f(x)-g(x) = 2x-5x = -3x$ while $(g - f)(x) = g(x) - f(x) = 5x - 2x = 3x$.

31. It is not associative. Let $f(x) = 2x, g(x) = 5x$, and $h(x) = 8x$. Then $[f - (g - h)](x) = f(x) - (g - h)(x) = f(x) - [g(x) - h(x)] = f(x) - g(x) + h(x) = 2x - 5x + 8x = 5x$, but $[(f - g) - h](x) = (f - g)(x) - h(x) = f(x) - g(x) - h(x) = 2x - 5x - 8x = -11x$.

32. It is commutative.
Proof: $(f \cdot g)(x) = f(x) \cdot g(x) = g(x) \cdot f(x) = (g \cdot f)(x)$ because multiplication in $\mathbb{R}$ is commutative.

33. It is associative.
Proof: $[(f \cdot g) \cdot h](x) = (f \cdot g)(x) \cdot h(x) = [f(x) \cdot g(x)] \cdot h(x) = f(x) \cdot [g(x) \cdot h(x)] = [f \cdot (g \cdot h)](x)$ because multiplication in $\mathbb{R}$ is associative.